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SUMMER SEMESTER-B.TECH (2026)
END SEMESTER EXAMINATION

Course Code: ICMTC04

Course Title: Mathematics III

Time: 3 Hours

Max. Marks: 50

Note: Attempt all questions. Attempt any two parts from each question. Missing data (if any) may be suitably assumed.

Q.No	Attempt all the five questions.	Marks	CO
1(a)	Using Cauchy Residue theorem, evaluate $\int_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)} dz$, where $C : z = 3$.	5	1
1(b)	If $f(z) = u + iv$ is an analytic function and $u - v = e^x(\cos y - \sin y)$, find $f(z)$.	5	1
1(c)	Using Cauchy Integral Formula, evaluate $\oint_C \frac{z}{z^2+1} dz$ where C is the circle: 1. $ z + \frac{1}{2} = 2$ 2. $ z + i = 1$.	5	1
2(a)	Apply Stoke's theorem to evaluate the integral $\oint_C (x+y)dx + (2x-z)dy + (y+z)dz$ over the closed curve C , where C is the boundary of the triangle with vertices $(2, 0, 0)$, $(0, 3, 0)$ and $(0, 0, 6)$.	5	2
2(b)	If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = \vec{r} $, show that $\nabla(r^n) = nr^{n-2}\vec{r}$ and $\nabla(\vec{a} \cdot \vec{r}) = \vec{a}$ where $\vec{a}$ is a constant vector.	5	2
2(c)	Evaluate $\iint_S \vec{A} \cdot \hat{n} dS$ where $\vec{A} = 18z\hat{i} - 12\hat{j} + 3y\hat{k}$ and S is the part of the plane $2x + 3y + 6z = 12$ which is in the first octant.	5	2
3(a)	Solve by Laplace Transform: $(D^2 + 2D + 5)y = e^{-t} \sin t$ given that $y(0) = 0$, $y'(0) = 1$.	5	3
3(b)	Find the Laplace transform of the function given by: $f(t) = \begin{cases} t, & 0 < t \leq c \\ 2c - t, & c < t < 2c \end{cases}$	5	3
3(c)	State Convolution theorem. Using it to evaluate $L^{-1} \left(\frac{s^2}{(s^2+a^2)(s^2+b^2)} \right)$.	5	3
4(a)	Show that the series $\frac{l}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \left(\frac{2n\pi x}{l} \right)$ represents $\frac{1}{2}(l-x)$ when $0 < x < l$.	5	4

Q.No	Attempt All the five questions.	Marks	CO																
4(b)	For $-\pi < x < \pi$, find Fourier series for the function $f(x) = x + x^2$. Hence show that $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$	5	4																
4(c)	Express $f(x)$ in Fourier series up to the third harmonic from the following table: <table><tr><td>x</td><td>0°</td><td>60°</td><td>120°</td><td>180°</td><td>240°</td><td>300°</td><td>360°</td></tr><tr><td>$f(x)$</td><td>1.98</td><td>2.15</td><td>2.77</td><td>-0.22</td><td>-0.31</td><td>1.43</td><td>1.98</td></tr></table>	x	0°	60°	120°	180°	240°	300°	360°	$f(x)$	1.98	2.15	2.77	-0.22	-0.31	1.43	1.98	5	4
x	0°	60°	120°	180°	240°	300°	360°												
$f(x)$	1.98	2.15	2.77	-0.22	-0.31	1.43	1.98												
5(a)	Find the solution of the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ which satisfies the following boundary conditions: (i) $u \rightarrow 0$ as $y \rightarrow \infty$ for all x , (ii) $u = 0$ at $x = 0$ for all y , (iii) $u = 0$ at $x = 1$ for all y , (iv) $u = x(1 - x)$ if $y = 0$ for all $x \in (0, 1)$.	5	5																
5(b)	Solve the heat equation $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$, if (i) $u(0, t) = 0$, (ii) $u_x(l, t) = 0$, (iii) $u(x, t)$ is bounded and (iv) $u(x, 0) = \frac{u_0 x}{l}$ for $0 \leq x \leq l$.	5	5																
5(c)	A string is attached and fastened to two points distance l apart and displaced into the form $u(x, 0) = 3(lx - x^2)$ from which it is released at time $t = 0$. Find the displacement $u(x, t)$ of the string at a distance x from one end.	5	5																